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Tunnel FET-based ultra-lightweight reconfigurable TRNG and PUF design for resource-constrained internet of things

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Summary

Conventional complementary metal oxide semiconductor (CMOS)-based dedicated true random number generator (TRNG) and physically unclonable function (PUF) designs have exhibited higher energy consumption and large area overhead with technology scaling. In contrast, this paper for the first time presents emerging tunnel field-effect transistor (TFET)-based ultra-lightweight reconfigurable TRNG and PUF design. A unique methodology is proposed that considers p-i-n forward current characteristics to generate random keys for PUF and TRNG designs. Leveraging the p-i-n forward current of TFET, a ring oscillator (RO) is designed that exhibits variation in the operating frequency and acts as the main source of entropy for both PUF and TRNG. The TRNG in the proposed design is robust and passed all the National Institute of Standards and Technology (NIST) tests. Considering the variation in p-i-n forward current, the uniqueness of PUF is calculated as 47.5% and 46% at a supply voltage of 0.5 and 0.45 V, respectively. Moreover, PUF achieves high reliability of 82% and 88.7% with supply voltage and temperature variations, respectively. The proposed design is proved to be compact with relatively lower gate count and shows low energy consumption of 4.8 pJ/bit at 0.5-V supply voltage. With these performance metrics, the proposed design is highly suitable for resource-constrained internet of things (IoT).

KEYWORDS

internet of things (IoT), physically unclonable function (PUF), p-i-n forward leakage, true random number generator (TRNG), tunnel field-effect transistor (TFET)

1 | INTRODUCTION

Hardware security is becoming an essential metric due to modern security attacks and serious vulnerabilities on electronics.[1] The divergent security attacks include hardware Trojan attacks, differential power analysis attacks, and hardware reverse engineering that leak confidential information of systems.[2] The hardware become more vulnerable due to deterministic cryptographic algorithms, utilizing pseudo random number generators, and insecure crypto-key storage for encryption.[3] On the other hand, true random number generator (TRNG) and physically unclonable function (PUF) are emerging as the critical security primitives to generate on-the-fly secure keys for modern cryptographic applications.[4] PUF and TRNG designs have been leveraged for divergent applications including E-health, Wi-Fi authentication

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protocols, patient monitoring, encryption–decryption, secure wireless communications, and information security.[5,6] On the other hand, light-weight security algorithms, primitives and methodologies have attracted recent attention due to the resource-constrained IoT nodes that are compact and operate with low-power batteries.[7] In support of this, many complementary metal oxide semiconductor (CMOS)-based TRNG and PUF designs have been proposed exploring oscillator (RO) jitter, oscillator collapse, flip-flop metastability, and comparator common-mode noise.[8,9] Recently, a unified PUF and TRNG concept has been demonstrated for applications like the privacy-preserving mutual authentication (PPMA) scheme that requires the simultaneous presence of both primitives at IoT edge devices.[10] However, CMOS-based hardware security primitives show many challenges specifically in terms of performance, area, energy consumption, and design complexity.[11–14]

An alternative to CMOS technology, emerging transistors aim to improve energy efficiency at lower channel lengths and exhibit higher speed.[15] In support of this, several nanoelectronic devices including tunnel field-effect transistor (TFET), hybrid phase transition FET (HyperFET), carbon nanotube FET (CNTFET), silicon nanowire FET (SiNWFET), symmetrical tunneling FET (SymFET), and spin-transfer torque magnetic tunnel junction (STT-MTJ) have been proposed to improve the performance with ultra-low energy consumption.[16–18] Besides, the two-dimensional materials-based dichalcogenide TFETs and graphene nanoribbon FETs have also show higher potential for energy efficient circuit design.[19,20] Additionally, the ferroelectric FETs leveraging two-dimensional phase transition material have exhibited lower subtheshold swing (<60 mV/dec).[21] Among the wide variety of emerging devices, TFET exhibits steep subthreshold slope values at lower supply voltages.[22] TFET devices work with band-to-band tunneling phenomenon and show high I_{ON}/I_{OFF} ratio. TFETs have proved as potential candidates for high speed and energy-efficient digital logic design and applications.[23] With high g_m/i_d values, TFETs-based analog and RF circuits/systems have also achieved improved gain and bandwidth.[24] Apart from the benefits, TFET exhibits several special characteristics including unidirectional current conduction, miller capacitance effect, ambipolarity, and p-i-n forward current that limit their utility for regular logic and memory applications.[25] Although post-CMOS devices cannot replace the mainstream role of CMOS, their special characteristics have motivated the researchers to explore them for novel applications.[26,27]

In support of this, several works explored TFET unique device characteristics to enhance the hardware security of circuits/systems. Exploring the onset of tunneling for both positive and negative gate bias, TFET compact polymorphic logic gates were designed that show high resilience to cloning attacks.[27] TFET-based current mode logic and adiabatic logic were proposed to design differential power analysis resistant crypto systems along with achieving ultra-low power consumption.[28,29] Recently, a basic PUF is designed by exploring inherent variations of TFET that shows higher uniqueness and reliability.[30] TFET shows large p-i-n forward current that leads to functional failure of TFET-based SRAM design.[31] Several techniques have been demonstrated to avoid this forward current for general circuit design.[32,33] Orthogonal to this, the p-i-n forward current of TFET is explored as favorable feature to enhance the hardware security of circuits/systems.[34–36] In the recent works, authors have proposed TFET p-i-n forward based TRNG design that shows higher resilience towards reverse engineering attacks.[34] Leveraging the p-i-n forward current, TFET-based differential power analysis resistant crypto circuits have been demonstrated that achieve lower area overhead.[35] In addition, TFET p-i-n forward current is utilized to modify the flip-flop functionality for device-level hardware obfuscation application. However, none of the works have proposed reconfigurable TRNG and PUF design and its analysis. Moreover, the recent works have not considered the impact of supply voltage and temperature variations on the design. In this new contribution, an ultra-lightweight reconfigurable TRNG and PUF design is demonstrated by considering supply voltage and temperature variations. The unique keys of PUF have been generated by exploiting the variations in the p-i-n forward current of TFET. The unified PUF and TRNG design is analyzed and tested through calculating several parameters. The rest of the paper is organized as follows. In Section 2, TFET device structure, models, and distinct characteristics of TFET are discussed. More specifically, this section illustrates the p-i-n forward characteristics of TFET and the unconventional behavior of TFET transmission gate (TG). Section 3 presents the design of TFET-based reconfigurable PUF/TRNG design and its evaluation. Further, a discussion has been presented by comparing the proposed design with existing literature. Finally, Section 4 offers conclusion to the work.

2 | TFET CHARACTERISTICS AND DESIGN METHODOLOGY

This section presents TFET device structures, models, and characteristics. The impact of forward current on TFET TG is also demonstrated. Additionally, the design methodology of reconfigurable PUF and TRNG is explained in this section.

2.1 | TFET device model and p-i-n characteristics

This work explores a double-gate InAs TFET model with a channel length of 20 nm for reconfigurable PUF and TRNG design.[37] This model is developed by the University of Notre Dame to perform TFET-based circuit design and simulations. Figure 1 shows the structures of n-channel TFET (NTFET) and p-channel TFET (PTFET) with two gates (top-gate and bottom-gate). The main device parameters including channel length ($L_G = 20$ nm), equivalent oxide thickness ($EOT = 0.2$ nm), and channel thickness ($t_{ch} = 5$ nm) have depicted in Figure 1. More details including TFET device dimensions, equivalent circuit model, doping information of different layers, and different junction capacitance information can be found in Hao et al. and Lu et al.[37,38] The model is implemented based on the Kane-Sze formula for tunneling and validated in all four operating quadrants. This InAs TFET mathematically models divergent characteristics like main tunneling current, ambipolar current, p-i-n forward current, and negative differential resistance (NDR). The drain current of TFET is modeled by the addition of three currents that can be expressed as 1

$$I_d = I_{dt} + I_{da} + I_{de}, \tag{1}$$

where I_{dt} is the main drain-source tunneling current, I_{da} is the ambipolar current, and I_{de} is the sum of forward-biased p-i-n diode current and tunnel diode current.

Figure 2 shows the distinct characteristics of InAs NTFET that are described in four quadrants as a function of gate-to-source voltage (V_{GS}) and gate-to-drain voltage (V_{DS}).[37] In the first quadrant, when $V_{GS} > 0$ and $V_{DS} > 0$, TFET switches ON with source-channel band-to-band tunneling mechanism. In the second quadrant, when $V_{GS} < 0$ and $V_{DS} > 0$, TFET exhibits ambipolar characteristics with the tunneling between channel to drain. In the third quadrant, when $V_{DS} < <0$ (large negative V_{DS}), p-i-n structure of NTFET works in forward bias condition. Consequently, NTFET works as a forward biased p-i-n diode[31,37] and demonstrates significant drain current. This current is characterized as

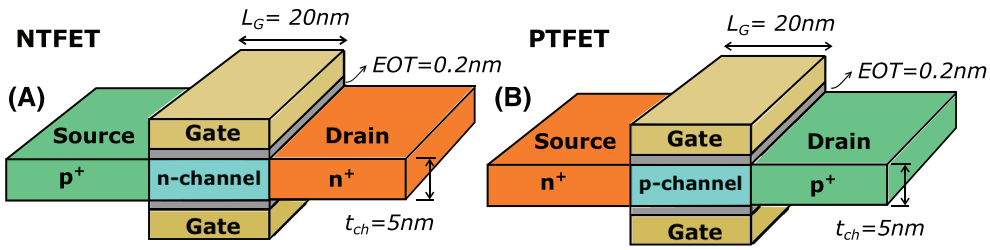

FIGURE 1 InAs double-gate TFET structure (A) NTFET (B) PTFET with main device parameters [Colour figure can be viewed at wileyonlinelibrary.com]

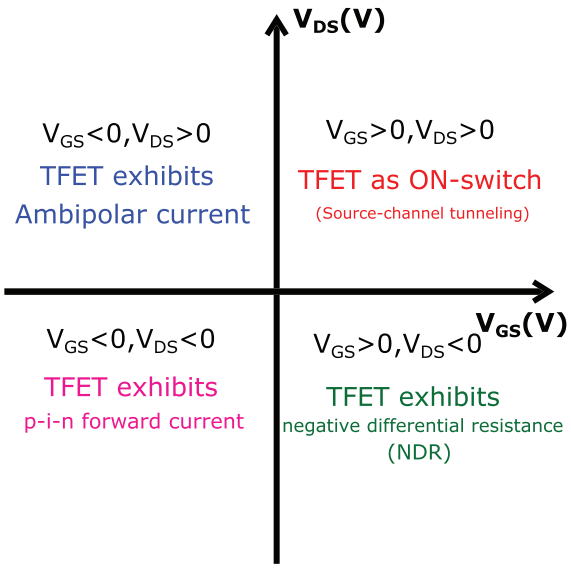

FIGURE 2 Distinct characteristics of NTFET in four different quadrants [Colour figure can be viewed at wileyonlinelibrary.com]

TFET p-i-n forward current. Recently, several works have explained about TFET p-i-n forward current and the phenomenon behind its occurrence.[31–33] Apart from this, in the fourth quadrant with $V_{GS} > 0$ and $V_{DS} < 0$, TFET works as tunnel diode and exhibits NDR characteristics.[31–33,37,38] This TFET model is popularly known as universal TFET model and explored by several researchers for the TFET circuit design.[37,38] In this work, TFET circuits have been implemented using Verilog-A netlist and simulated exploring industry standard Synopsys h-spice environment.

Figure 3A shows the simulated I_D–V_{DS} characteristics of NTFET with varying V_{DS} from −0.5 V to 0.5 V. It can be depicted that unlike MOSFET, TFET shows asymmetrical characteristics with varying V_{DS}. TFET with band-to-band tunneling exhibits large ON current at $V_{DS} > 0$, and $V_{GS} > 0$. In contrast, at low negative V_{DS} ($V_{DS} < 0$), TFET shows smaller drain current. Additionally, this current increases exponentially at large negative V_{DS} ($V_{DS} < <0$). This large drain current at increased negative V_{DS} bias, referred as p-i-n forward current. At large negative V_{DS}, TFET p-i-n structure enters into forward bias and gate losses the control of channel.[31] With heavily doped p-type, n-type regions and at large negative V_{DS}, the TFET p-i-n current flows from source to drain. TFET demonstrates approximately 20-μA/μm forward current at $V_{DS} = -0.5$ V. Figure 3B shows the I_D–V_{DS} characteristics with log scale on y-axis and demonstrates the identical forward current with variation in V_{GS} bias. Specifically, at divergent V_{GS} voltages (0.1 to 0.5 V), TFET exhibits similar forward current. The reason is that at large negative V_{DS} bias, this current is hardly controlled by the gate voltage. Several works have explained the concept of TFET forward current and demonstrated it experimentally.[31–33] Table 1 compares the divergent TFET designs and the reported p-i-n forward current in the literature. It shows that TFET demonstrates significant forward current and it is relatively lower compared to ON current. On the other hand, in fourth quadrant ($V_{DS} < 0$, and $V_{GS} > 0$), TFET works as tunnel diode with the band-to-band tunneling from drain to channel.[31,32,37] In this region, at lower negative V_{DS} (at $V_{DS} = -0.1$ V), TFET drain current is reduced with an increase in V_{DS} and exhibits NDR characteristics. At large negative V_{DS}, TFET exhibits significant forward current because with increase in negative V_{DS}, the V_{GS} reduces (become $V_{GS} < 0$). Consequently, p-i-n structure with $V_{DS} < 0$ and $V_{GS} < 0$

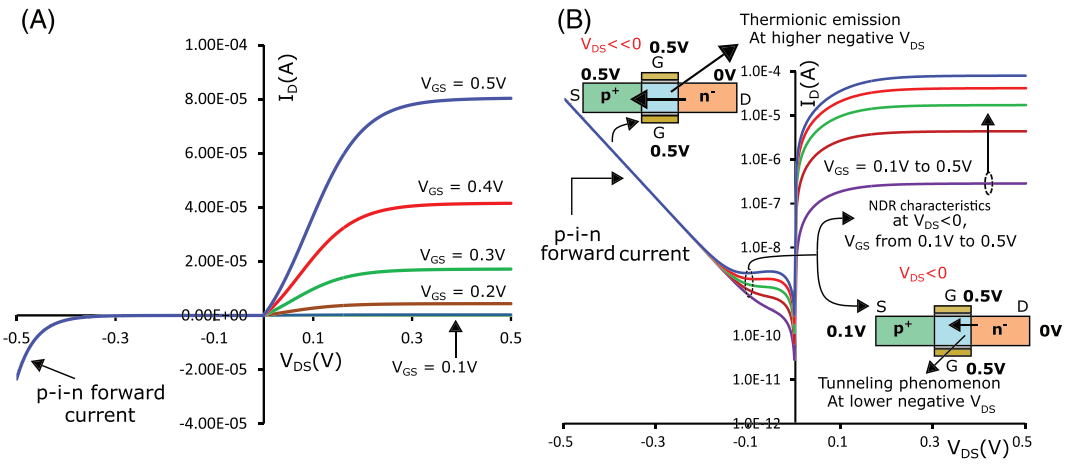

FIGURE 3 InAs NTFET (A) I_D–V_{DS} characteristics (B) I_D–V_{DS} characteristics with log scale on y-axis by varying V_{DS} from −0.5 to 0.5 V [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Comparison of TFET devices ON current with p-i-n forward current

Parameter	22	31	37	39
Type	Experimental	Experimental	Analytical Verilog-A model (explored in this work)	LUT-based Verilog-A model
Device	InAs/GaAsSb/GaSb TFET	InAs/Si TFET	TFET	InAs TFET
Channel length (nm)	100	250	20	20
I_{ON} (μA/μm)	92	5	100	147
p-i-n forward current (μA/μm)	2	1	20	60

exhibits large p-i-n current.[37,38] It can be concluded that TFET at lower negative V_{DS} ($V_{DS} < 0$, $V_{GS} > 0$) acts as tunnel diode and exhibits NDR characteristics. On the other hand, at large negative V_{DS} ($V_{DS} < <0$), TFET forward biased p-i-n diode shows significant forward current that cannot be controlled by gate voltage. This significant forward current is explored in TFET based reconfigurable TRNG and PUF designs.

2.2 | Design procedure

With forward biasing of TFET p-i-n architecture (with negative V_{DS}), the TFET acts as p-i-n diode and exhibits a significant forward current. TFET inherently exhibits higher random dopant fluctuations, geometric variations, and gate work function variations.[30,40–42] Moreover, with scaled channel lengths, TFET exhibits large random dopant fluctuations due to the formation of ultra-sharp doping profile and diffusion of charge at junction regions.[40–42] Due to these inherent changes, manufactured TFET IC exhibits random variations in device characteristics. As a result, TFET inherently exhibits random p-i-n forward current, and separate doping variations are not required. To obtain these variations (inherently results from device parameters) with the analytical TFET model, this work introduces a parameter "*K*". The p-i-n forward current is defined as 2

$$I_{p-i-n} = K \times I(-V_{DS}), \tag{2}$$

where $I_{p\text{-}i\text{-}n}$ is modified p-i-n forward current for circuit design and "*K*" is a parameter chosen to get variation in its magnitude. The parameter "*K*" is introduced to obtain the variations in forward current that is explored to test PUF functionality (like uniqueness). Figure 4 depicts the I_D–V_{DS} characteristics that shows different magnitudes of p-i-n forward current by varying the value of "*K*". As this current is lower compared to ON current,[22] the value of "*K*" is chosen to obtain the p-i-n forward current of 0.25×, 0.5×, and 0.75× the ON current. A little variation in "*K*" (naturally generated with uncontrollable geometric, dopant, and gate work function variations in TFET) exhibits a difference in magnitude of p-i-n forward current that changes the response of reconfigurable TRNG and PUF design. With the resulted variation in forward current, the following steps need to be performed for designing a reconfigurable PUF and TRNG:

1. Firstly, a random value of forward current is generated (by selecting "*K*" value in the model) that is leveraged in designing reconfigurable TRNG and PUF.
2. Later, TFET-based circuits have been developed exploring p-i-n forward current. In this work, TFET TG-based RO is designed and explored as a common entropy source. The functionality of RO relies on the TFET p-i-n forward current. As a result, attackers cannot easily anticipate the behavior of TFET circuits.
3. Finally, The RO design is exploited as a common source of entropy for reconfigurable PUF and TRNG design.

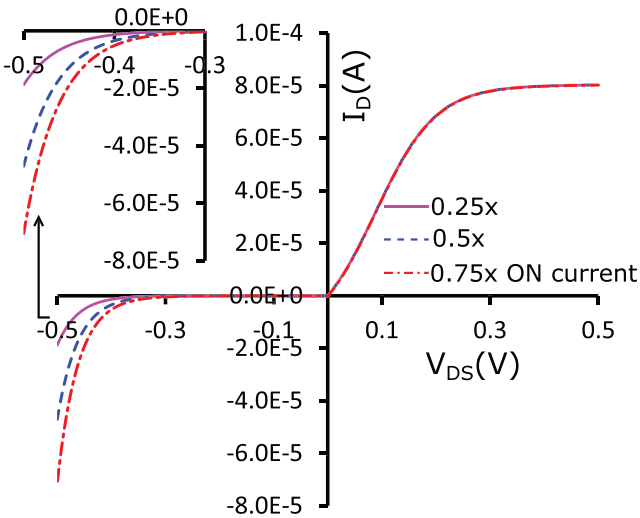

FIGURE 4 I_D–V_{DS} characteristics of NTFET demonstrating different magnitudes of p-i-n forward current with variation in parameter "*K*" [Colour figure can be viewed at wileyonlinelibrary.com]

2.3 | Behavior of TFET TG

TFET TG shows unconventional behavior because of p-i-n forward current and this can be utilized in reconfigurable TRNG and PUF design. Consider a TFET TG design that consists of an input (V_{in}), the output (V_{out}) and driving a load capacitor (C_L) as shown in Figure 5A. An enable (*En*) and its compliment (En_b) are used to switch ON/OFF the TFET TG design. With active low (logic "0") *En* signal, the TFET TG turns ON. On the other hand, TFET TG is supposed to be turned OFF with the active high (logic "1") *En* signal. Contrast to this, TFET TG partially switches ON because of significant p-i-n forward current. As a result, TFET TG shows high resistance with active high *En* and this region is named as asymmetrical region. The identical functionality can be observed while simulating InAs TFET TG design at a supply voltage of 0.5 V. With active high *En* signal, TFET TG partially turns ON, and the output results according to the applied input as shown in Figure 5B. This is due to the fact that with the active high enable signal, TFET in TG design shows significant p-i-n forward current and helps to charge/discharge the output of TG with higher propagation delay. As a result, TFET TG exhibits unconventional behavior that is explored for reconfigurable TRNG and PUF design.

It is identified that the TFET TG designs (with p-i-n forward current) connected in cascaded chain fashion exhibits non-linear propagation delay. For example, consider a TFET TG-based cascaded chain that consists of four-stages as shown in Figure 6A. This delay line comprises input (V_{in}), output (V_{out}), and the control bits (X_1 to X_4) to vary the propagation delay. When active low enable signal is applied, TFET TG switches ON and exhibits lower propagation delay. On the other hand, with active high enable, TFET TG due to p-i-n forward current exhibits higher propagation delay. Considering this effect, the TFET TG delay chain demonstrates non-linear delay values by varying the control bit combinations as shown in Figure 6B. It can be observed that propagation delay of cascaded chain increases by increasing the number of TGs in asymmetrical region (TG with active high enable). Additionally, the propagation delay increment is relatively high with successive TGs in the asymmetrical region (successive TG designs with active high enable). The variation in propagation delay is observed with a little modification in control bit pattern wherein fixed number of TGs and successive TGs are in asymmetrical region. With these characteristics, the cascaded chain when connected as an RO exhibits an unpredictable frequency of oscillations that acts as a common entropy source in proposed reconfigurable PUF and TRNG design.

3 | PROPOSED DESIGN AND ANALYSIS

This section presents the architecture of reconfigurable TRNG and PUF design. The proposed design is evaluated by considering supply voltage and temperature variations. Further, a discussion has been presented by comparing the proposed design with existing literature.[8–10,34,43–45]

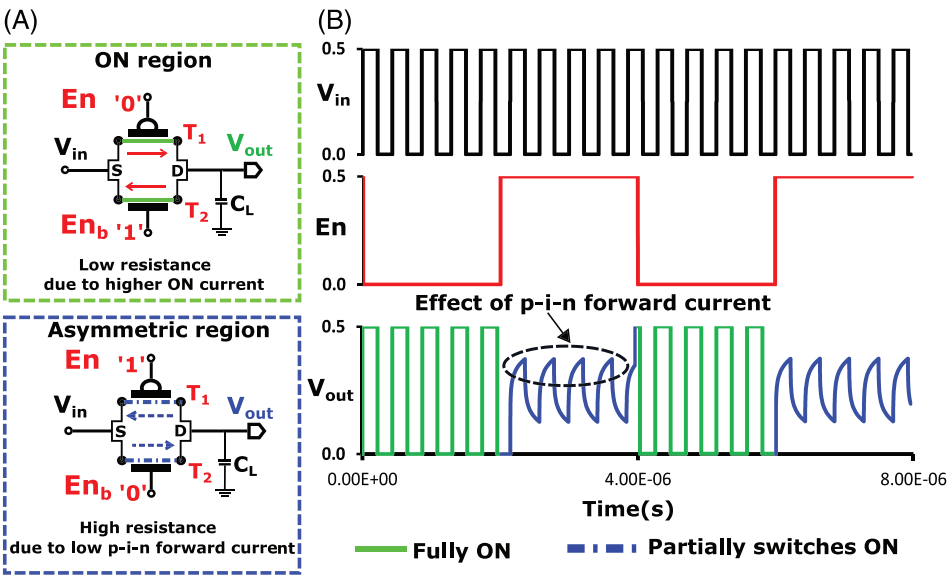

FIGURE 5 TFET TG (A) ON and asymmetrical switching mechanism (B) simulated transient response at 0.5-V supply voltage [Colour figure can be viewed at wileyonlinelibrary.com]

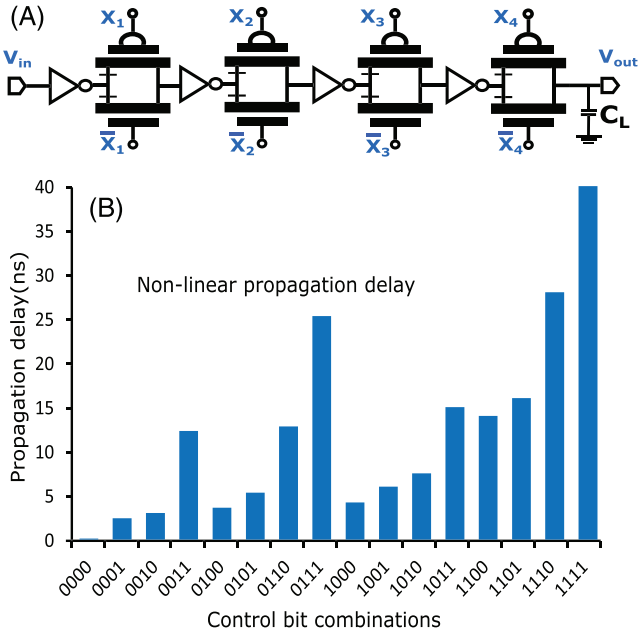

FIGURE 6 (A) TFET TG delay chain for 4-bit (B) propagation delay against different control bit combinations [Colour figure can be viewed at wileyonlinelibrary.com]

3.1 | Design

The architecture of proposed TFET-based reconfigurable PUF and TRNG design is as shown in Figure 7A. The TFET TG-based RO is proposed that acts as a common entropy source for reconfigurable PUF and TRNG design. A TFET 11--stage RO is designed wherein 10 TGs have been embedded between inverter stages as shown in Figure 4A. The control bits X_0, X_1, X_2 ... ... and X_9 (X[0:9]) are used to vary the propagation delay of TFET TGs in the RO design. As discussed in Section 2.3, the TFET TG-cascaded chain resulted arbitrary delay values by varying control bits. Consequently, the designed TFET TG-based RO demonstrates variation in operating frequency. The characteristics and detailed analysis of RO have been presented in the prior works.[34,35] Apart from this, a 10-bit counter is explored to record the oscillations of TFET RO. To update the output of reconfigurable TRNG and PUF design in every clock (*CLK*) cycle, a 10-bit register is also used. Selection line "*S*" is responsible to operate the proposed design either as PUF or TRNG with the help of multiplexers as shown in Figure 7A.

The functionality of the reconfigurable PUF and TRNG design is explained by using the transient response as shown in Figure 7B. When "*S*" is at logic "0", upon the rising of the *CLK* signal, RO exhibits oscillations. In this configuration, the proposed design acts as PUF, the frequency of RO depends on the external control bits (P[0:9]) applied through the multiplexers. A 10-bit counter is utilized to record the oscillations of RO. At the falling edge of the *CLK*, 10-bit register updates to the value produced by counter and the counter resets. Thus, PUF produces the 10-bit response (R_{OUT} [0:9]). When "*S*" is at logic "1", at the next rising edge of the *CLK* signal, RO again exhibits oscillations. The frequency of RO depends on the control bits (X[0:9]). In this configuration, the design appears to be a continuous loop and hence acts as a TRNG. The counter records the RO oscillations and at the falling edge of the *CLK*, 10-bit register updates the count. Thus, TRNG produces the 10-bit response (R_{OUT} [0:9]). The design explores the response (R_{OUT} [0:9]) produced in the present *CLK* as RO control bits (X [0:9]) in the next *CLK* cycle for TRNG. Therefore, the design allows continuous change in control bit combinations that produces the divergent TRNG responses.

3.2 | Analysis

TFET reconfigurable PUF and TRNG design has been tested by calculating important metrics. The performance of PUF is evaluated by using average interchip hamming distance (HD), and it is also named as uniqueness. The uniqueness of PUF[5] is defined as Equation 3

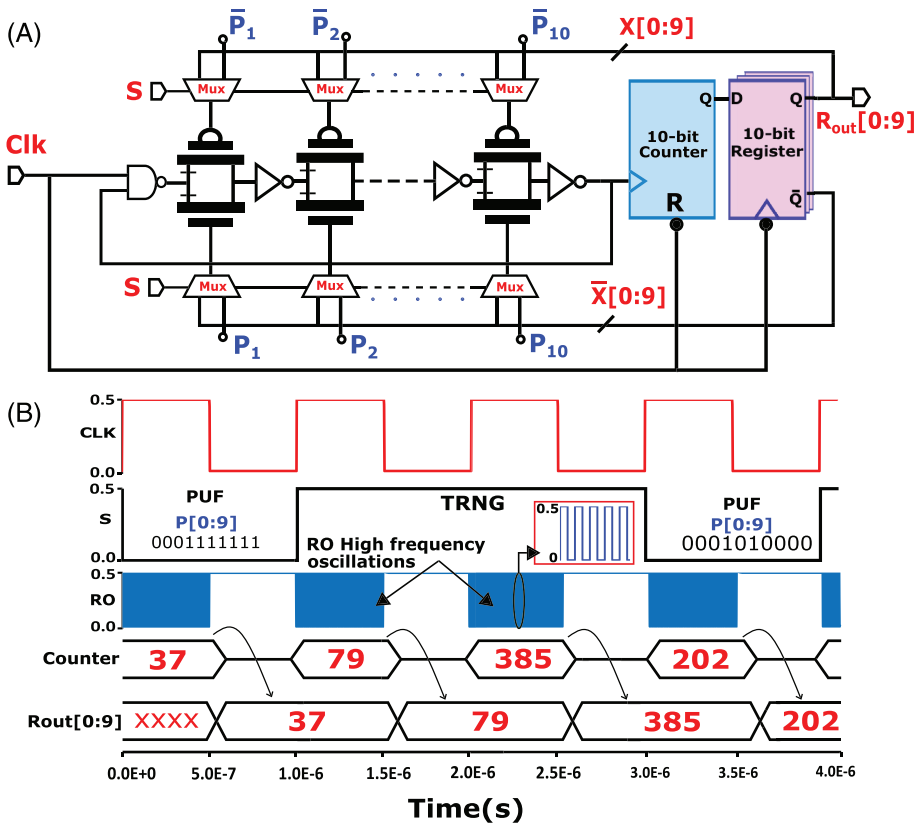

FIGURE 7 Proposed TFET based reconfigurable PUF and TRNG (A) circuit design (B) transient characteristics [Colour figure can be viewed at wileyonlinelibrary.com]

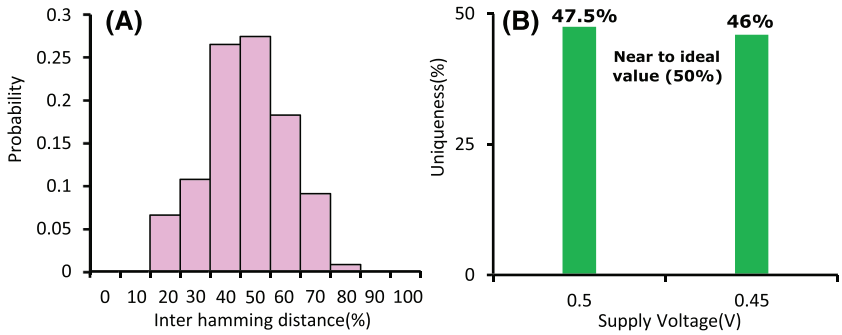

FIGURE 8 Proposed PUF (A) interchip HD distribution (B) uniqueness with varying supply voltage [Colour figure can be viewed at wileyonlinelibrary.com]

$$Uniqueness = \frac{2}{n \times (n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{HD(R_i, R_j)}{N}, \tag{3}$$

where n is the number of sampled response strings and N is the bit-length of a response string, (R_i, R_j) are the two output strings from different PUF designs, and HD (R_i, R_j) is the hamming distance between two binary strings. Uniqueness or the average interchip HD of the proposed design is calculated by applying 32 random inputs to five different PUFs. These five different PUF circuits are designed by varying the p-i-n forward current in the range of 0.25× to 0.75× ON current. Figure 8A shows the interchip HD of proposed design which is observed to be a normal distribution. It can be observed that the maximum and minimum values of normal distribution are 20% and 80%, respectively. The average interchip HD (uniqueness) of the proposed design is resulted as 46% and 47.5% at supply voltages of 0.45 and 0.5 V,

respectively, as shown in Figure 8B. From this, it can be shown that the proposed PUF exhibits high uniqueness at different supply voltages. Thus, the proposed reconfigurable design as PUF achieves high uniqueness (near to ideal value of 50%).

Another important metric is average intra-chip hamming distance (HD), and it is referred as the reliability of PUF. The reliability of PUF[5] is defined as 4

$$Reliability = 1 - \frac{1}{n}\sum_{j=1}^{k}\frac{HD\left(R_i, R_{i,j}\right)}{l}, \tag{4}$$

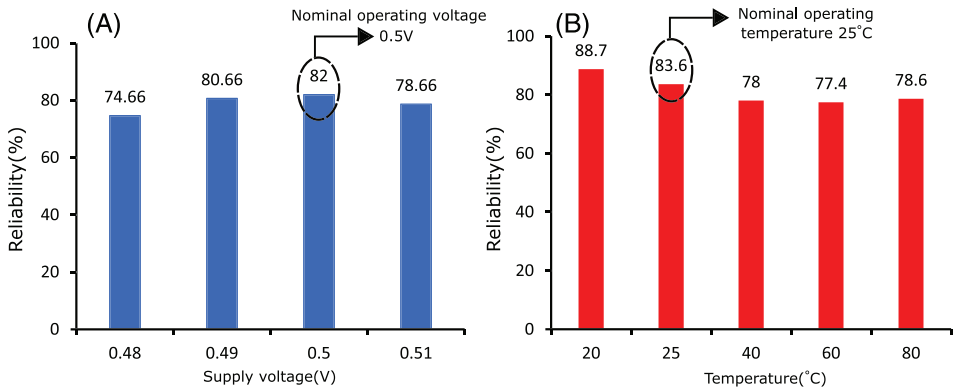

FIGURE 9 Proposed PUF reliability with (A) varying supply voltage at 25°C temperature (B) varying temperature at 0.5-V supply voltage [Colour figure can be viewed at wileyonlinelibrary.com]

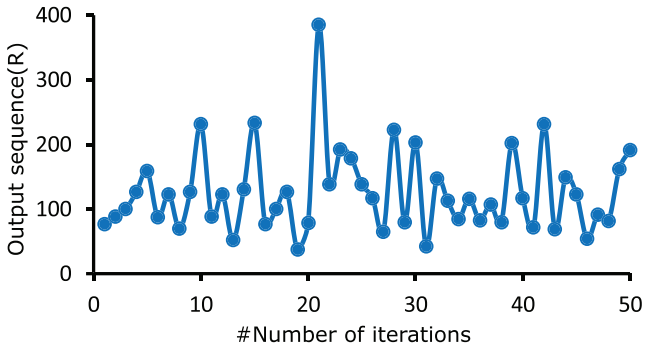

FIGURE 10 Output sequence of proposed TRNG obtained from 50 clock cycles [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 TFET TRNG NIST test results with 100 kb sequence

NIST test[a]	Average p value	Proportion (out of 55)	p_T value
Frequency	0.47	52	0.02
Block frequency	0.374	54	0.001
Approximate entropy	0.271	52	0.023
Runs	0.569	55	0.34
Longest runs of 1's	0.411	53	0.65
Cumulative sums-1	0.2	52	0.088
Cumulative sums-2	0.35	54	0.001
Serial-1	0.32	52	0.001
Serial-2	0.43	54	0.006
Binary matrix rank	0.42	52	0.05
DFT	0.32	52	0.003

[a]Minimum p_T value is 0.0001 and minimum pass rate is 52/55 according to NIST Publication 800-22 Revision 1.[46]

TABLE 3 Performance comparison of proposed design with existing literature

Parameter	8	9	43	44	45	10	34	This work
Method	Tetrahedral oscillator	Oscillator collapse	Oscillator collapse	Oscillator jitter	FinFET metastability	FinFET metastability	TFET p-i-n forward current	TFET p-i-n forward current
Design	TRNG	PUF	TRNG	TRNG	TRNG	Unified TRNG/PUF	TRNG design	Reconfigurable TRNG/PUF
Technology (nm)	130	40	40	28	14	14	20	20
V_{DD} (V)	1.8	0.5	0.9	0.9	0.5	0.65	0.5	0.5
Power consumption (μW)	40	28.4	46	540	250	72.54	54	48
Energy consumption (pJ/bit)	400	17.75	23	23	5	7.25	5.4	4.8
Gate equivalent[a]	1033	2517	2458	502	4900	1181	388	407
Bitrate (Mb/s)	0.1	1.6	2	23.16	50	1480	10	10
Variation effect (temperature, voltage)	Considered	Considered	Considered	Considered	Considered	Considered	Not considered	Considered

[a]Gate equivalent is calculated by standardizing all TRNG designs to 10-bit and considered the static inverter as reference gate.

where n is the number of input strings applied, and l is the bit-length of a PUF response. HD (R_i, $R_{i,j}$) is the hamming distance, R_i is the response of a PUF at nominal operating conditions, and $R_{i,j}$ is the response of the same chip at the different environmental conditions. The reliability is calculated by applying 32 random inputs to proposed PUF design, under different environmental conditions. A supply voltage of 0.5 V and a room temperature of 25°C is considered as nominal operating conditions. Figure 9A presents the reliability of proposed PUF with a supply voltage variation from 0.48 to 0.51 V. From this, it can be shown that the proposed PUF design achieves maximum reliability of 82%. Figure 9B shows the reliability of PUF by varying the temperature from 20°C to 80°C. It can be depicted that TFET PUF exhibits a maximum reliability of 88.7%. Thus, the proposed PUF design exhibits higher reliability under different environmental conditions. In addition, the bitstream quality of the proposed design when working as TRNG is also tested. The generated output sequence of proposed TRNG is shown in Figure 10. It can be depicted that the generated output sequence is observed to be random. Moreover, divergent NIST tests have been performed on output sequence using NIST Pub RNG 800-22 Revision 1.[46] Table 2 shows the result of various NIST tests performed on proposed TRNG design with 100-kb sequence. From the results, it can be observed that the proposed TRNG with uncorrelated bit stream has passed all the NIST tests performed.

Proposed TFET-based reconfigurable PUF and TRNG design is implemented using InAs TFET at 20-nm technology. This design is compared with existing CMOS designs as shown in Table 3. As the proposed design explores TFET RO jitter as a source of randomness, it has been compared with several CMOS RO-based TRNG and PUF design. Several performance parameters including power consumption, energy consumption, bitrate, and gate equivalent (or count) of TRNG designs have been compared. The gate equivalent of the TRNG design is calculated by following Nikonov and Young.[47] In Nikonov and Young,[47] it has been proved that both TFET inverter and CMOS inverter have shown identical area. Henceforth, this work considers inverter as reference gate for gate equivalent calculation. The proposed design exhibits a lower gate equivalent of 407. The lower gate count is resulted by exploring special characteristics (p-i-n forward current) of TFET. On the other hand, CMOS-based TRNG/PUF designs exploring tetrahedral RO, even stage oscillator collapse, 3-edge multi-mode RO, and metastability have achieved higher gate count as shown in Table 3. Additionally, the proposed PUF design is analyzed by considering the temperature and supply voltage variations. Under these variations effect, proposed design achieves a higher uniqueness (47.5%) and reliability (88.7%). Furthermore, the proposed design achieves a low energy consumption of 4.8 pJ/bit with a bit rate of 10 Mb/s. Thus, the TFET unique characteristics made the proposed TRNG and PUF design ultra-lightweight and also achieve lower energy consumption.

4 | CONCLUSION

TFET-based reconfigurable TRNG and PUF design is demonstrated that leverages the special p-i-n forward characteristics of the device. This unique behavior encouragingly leads to unpredictable frequency of oscillations in TFET RO design that act as a main source of entropy for both PUF and TRNG. The proposed TRNG passed all the NIST tests at a supply voltage of 0.5 V. The PUF is designed exploring a unique methodology that considers variation in TFET p-i-n forward current. At lower supply voltages of 0.5 and 0.45 V, the proposed PUF has shown a higher degree of uniqueness 47.5% and 46%, respectively. Further, PUF exhibited a high reliability of 82% and 88.7% with supply voltage and temperature variations, respectively. The proposed design demonstrated lower gate count with an energy consumption of 4.8 pJ/bit at 0.5 V. Thus, the proposed design achieves lower area overhead compared to the existing designs and highly suitable for resource-constrained IoT.

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